BLA-06 · NETWORKING AND STORAGE

Getting Requests In, and Keeping Data Close

Amazon CloudFront · Elastic Load Balancing · AWS Global Accelerator · Amazon Route 53 · Amazon EBS and EFS

PART 02

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Amazon CloudFront

A content delivery network — copies of your content at edge locations worldwide.

- 📍 Origin: an S3 bucket, a load balancer, or any web server
- 🔒 Origin Access Control: only CloudFront can read the S3 bucket

Content	Cache it?
Website pages, images, scripts	✅ Yes
An answer that's the same for everyone (FAQ)	✅ For a short time
A personal chat answer	❌ Never — pass it through

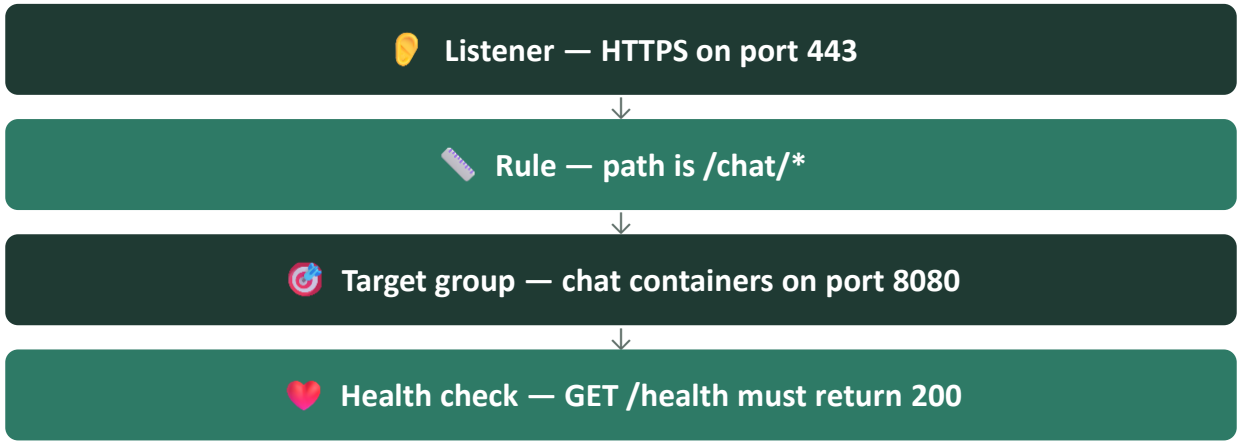
Elastic Load Balancing

Spreads requests across servers — and stops sending to the ones that fail.

Type	Works with	Use it for
Application LB	HTTP and HTTPS — reads the URL	Web apps and APIs; route by path or host
Network LB	TCP and UDP — very fast	Huge volumes, fixed IP addresses
Gateway LB	Network traffic	Sending traffic through firewalls

💔 **Health checks:** the load balancer only sends traffic to targets that pass.

How an ALB routes a request



⚠️ **LLM answers can be slow.** The ALB's idle timeout is 60 seconds by default — raise it, or stream the answer.

AWS Global Accelerator

Same edge locations as CloudFront — a different job.

	CloudFront	Global Accelerator
Does	Caches content at the edge	Routes traffic over the AWS network — no caching
Traffic	HTTP and HTTPS	Any TCP or UDP
Entry point	A domain name	Two fixed IP addresses
Best for	Websites, images, APIs	Voice, gaming, fast failover between regions

Amazon Route 53

DNS that can choose which answer to give.

Routing policy	What it does	Example
Simple	One answer	A single website
Weighted	Split by percentage	90% old version, 10% new model
Latency	Fastest region for the user	US users → us-east-1
Failover	Backup when the main is unhealthy	A standby region
Geolocation	Answer by the user's country	EU users stay in the EU

💡 **Alias records** point a domain straight at a load balancer, CloudFront or S3 — and those lookups are free.

Amazon EBS

A network drive for one EC2 instance, in one Availability Zone.

- 📸 Snapshots back it up — and copy it to another zone or region

Volume type	Good for
gp3 — general SSD	Most workloads
io2 — fastest SSD	Busy databases
st1 / sc1 — hard disk	Large files read in order, low cost

⚠️ **Terminating an instance deletes its root volume —** but extra volumes stay, and keep costing money.

Amazon EFS vs Amazon EBS

	EBS	EFS
Connects to	One instance	Many instances at once
Scope	One Availability Zone	Many Availability Zones
Size	You choose it — and pay for it	Grows and shrinks — pay for what you store
Works with	Linux and Windows	Linux (NFS)
Cost	Cheaper per GB	More per GB — Infrequent Access lowers it

🏢 **Several GPU instances share one EFS folder of model files —** download once, use everywhere.

Key takeaways

1. CloudFront caches content — never personal answers
2. ALB for web apps — and get the health check right
3. Global Accelerator: fixed IPs and fast failover, no caching
4. Route 53 weighted routing for gradual rollouts
5. EBS for one server, EFS for many

Next · Part 3: GenAI architecture trade-offs